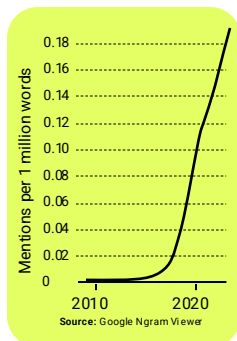
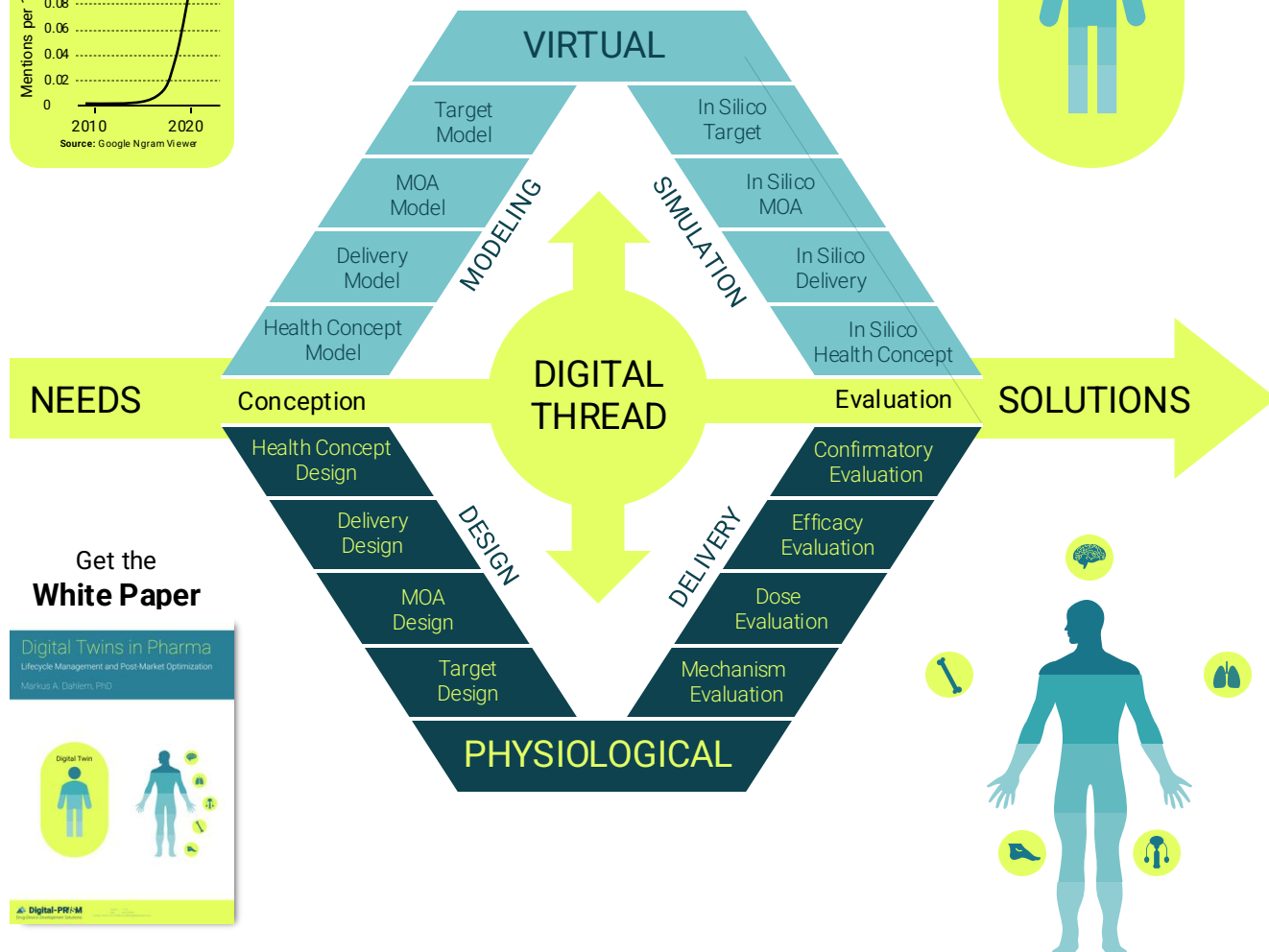


Digital Twin takes off



Digital Twin

Virtual patient



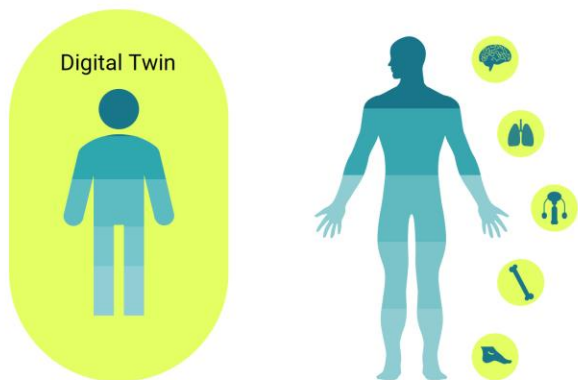
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Digital Twins in Pharma

Lifecycle Management and Post-Market Optimization

Markus A. Dahlem, PhD



Executive Summary

Everyone's excited about digital twins in healthcare – but not every model with real-time data is a digital twin. There is confusion what a digital twin actually is. Originally developed in engineering to optimize machines and processes, they are now positioned to transform medicine – by modeling not only physical structures, but also physiological systems, therapeutic interventions, health outcomes, and even entire patient journeys.

What defines a digital twin is not its complexity, but its capacity for control. A digital twin is dynamically coupled to its physical counterpart through continuous data exchange and evolves across its full lifecycle. This finally enables what precision medicine has long promised: care that is adaptive, feedback-driven, and truly individualized.

It is actually simple, in medicine, digital twins serve three core purposes: maintaining health, restoring it, and inventing new ways to do both. Their strength lies in replicating physiological and behavioral processes, synchronize with their human counterpart via data assimilation and then link gained insights with intervention design and real-world validation – through what we call *twin control*: the dynamic, bidirectional coupling of model and patient across the full lifecycle of care and therapy.

To organize the models and feedback loops that constitute twin control, we adapt the *Design Diamond* from industrial manufacturing to drug-device development: a lifecycle V-model and its mirrored virtual counterpart, connected by a digital thread. This structure doesn't just align with the pharmaceutical pipeline – it upgrades it.

Digital twins transform therapeutic products into learning systems. While early R&D and clinical trial applications remain limited by regulatory inertia, their first major impact is likely post-market – where real-world data can be assimilated to refine dosing, personalize care, and deliver differentiation, even in saturated therapeutic classes. From novel therapies to generics, digital twins enable continuous optimization.

Their promise is clear: intelligent therapeutics.

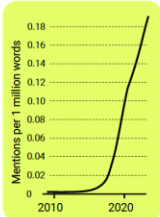
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The Rise & Promise of Digital Twins

Take Control.

The concept of the *digital twin* has evolved rapidly from a niche technical term to a foundational paradigm in modern system engineering and data-driven design. Although the term was coined in the early 2000s—most notably by Michael Grieves in the context of Product Lifecycle Management (PLM)—it remained relatively obscure for more than a decade.

That changed decisively in the mid-2010s.



Technological Drivers Enabling Digital Twins

- IoT & sensor technology
- Cloud Computing
- Big data analytics (AI/ML)
- Immersive Interfaces: AR and VR

IoT: Internet of Things; AI: Artificial Intelligence; ML: Machine learning things; AR: Augmented Reality; VR: Virtual Reality.

Figure Relative frequency of the term "digital twin" in published texts (Google Ngram). A rise from zero mentions to 0.2 in a million words in five years signals rapid conceptual adoption.

This rise in attention reflects a deeper shift:

From models that merely represent reality to models that interact.

The digital twin is not just a refined simulation—it is a synchronized virtual counterpart of a physical system, capable of adapting in real time and informing decisions through embedded intelligence.

A Digital Twin Promises Control.

2

Core Principles of Digital Twins

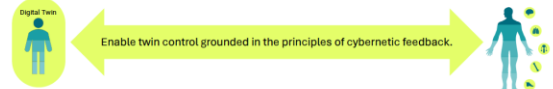
Not Every Model is a Twin.

Digital twins in medicine serve three principal purposes:

- Keeping healthy individuals healthy,
- Restoring health in ill patients,
- Developing novel therapeutics, such as drugs and medical devices.

Yet the term *digital twin* is often misunderstood.

Not every model, simulation, or dashboard with these aims qualifies. What defines a digital twin is its ability to synchronize dynamically with the real world, to evolve with it, and to generate insights or actions in response.



In industrial domains, twin control¹ is designed into the system across its lifecycle. The system being modeled—an engine, a turbine, a production line—is itself a product of engineering. In medicine, by contrast, the system is not designed but discovered: the human body, its pathologies, and the mechanisms of action we seek to influence all emerge from biology, not blueprint.

Even so, we retain the terminology. While **the lifecycle in this case is life itself** the underlying logic holds. A medical digital twin supports twin control through continuous sensing, interpretation, and adaptation—merging the lifecycle of the therapy with that of the patient, in an architecture that is cybernetic in spirit, if not in name.

This rests on two foundational features: **bidirectional coupling with the real world** and **lifecycle-spanning aggregation**. Together, they make twin control possible.

Twin is Lifecycle-spanning Bidirectional Coupling.

¹The term twin control was notably used in the Horizon 2020 project Twin-Control: *Twin Models for Simulation and Control of Machining Processes* (2015–2019), which applied digital twin technology to optimize industrial machine tools.

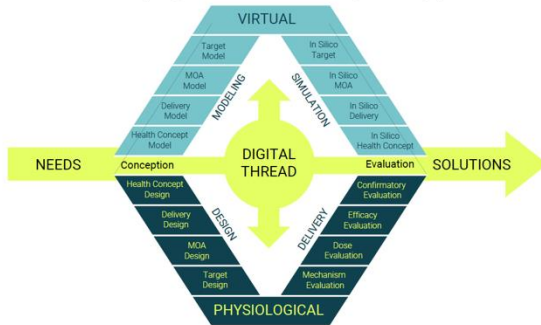
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The Design Diamond

From Intervention Design to Digital Control

The classical V-model maps the major steps in designing and delivering systems. As digital modeling and data integration evolved, this framework was mirrored—forming the **Design Diamond**.

Its halves are connected by a **Digital Thread**: data backbone linking models the physical to the virtual.



Pharmaceutical pipeline phases are mapped into the V-model—not merely for alignment with Product Lifecycle Management, but because the digital twin transforms pharmaceutical interventions into drug-device systems, introducing **Software Lifecycle Management** into the drug development process. This opens the door to iterative improvement of therapeutic products of truly intelligent health technologies.

Digital Control to Truly Intelligent Health Technologies

Contact

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*Let's explore how digital twins can advance your R&D strategy,
accelerate innovation, and enable truly intelligent health technologies.*